

Exploratory Data Analysis: Customer and Segment Patterns

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```
# Load packages needed for the future EDA report.
knitr::opts_chunk$set(
  echo = FALSE,
  warning = FALSE,
  message = FALSE
)
```

Introduction

Dataset Overview

Customer Profile Analysis

Customer Age

Customer age is explored as a foundational customer-profile characteristic. This descriptive comparison examines whether age distributions differ between fraud and non-fraud transactions in the balanced analysis sample; it does not imply that customer age causes fraud.

Table 1: Customer-age summary by fraud status

fraud_status	sample_size	mean_age	median_age	standard_deviation	minimum_age	maximum_age
Non-fraud	1000	42.19	42	13.47	18	83
Fraud	1000	41.85	42	13.35	18	85

The following faceted histogram provides a visual comparison of customer-age distributions between the two fraud-status groups.

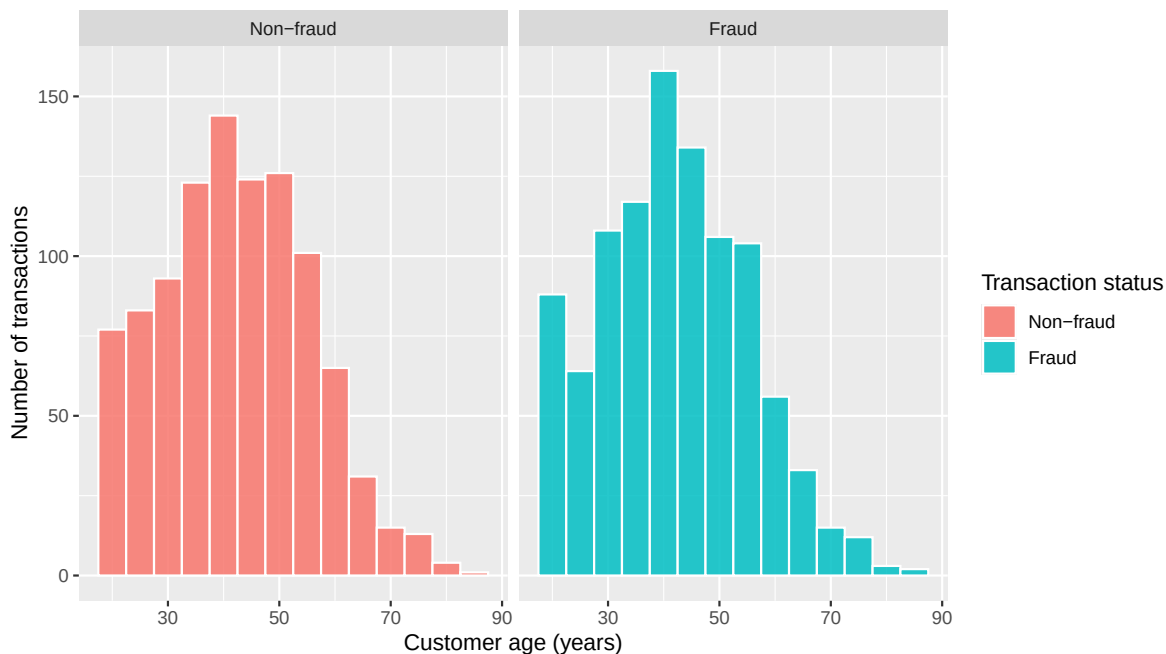


Figure 1: Distribution of customer age by fraud status in the balanced analysis sample.

The figure compares customer-age distributions between fraud and non-fraud groups in a sample balanced by design. It is intended only for descriptive comparison and should not be used to draw causal conclusions or estimate population fraud prevalence.

Credit Score

Credit score is examined as a customer financial-profile characteristic. Comparing credit-score distributions across fraud-status groups provides a descriptive view of whether fraudulent and non-fraudulent transactions are associated with different credit-score patterns in the balanced analysis sample.

Table 2: Credit-score summary by fraud status

fraud_status	sample_size	mean_score	med_score	std_score	min_score	max_score
Non-fraud	1000	677.20	679	79.66	405	850
Fraud	1000	679.16	682	79.89	418	850

The following faceted histograms compare credit-score distributions between the two fraud-status groups.

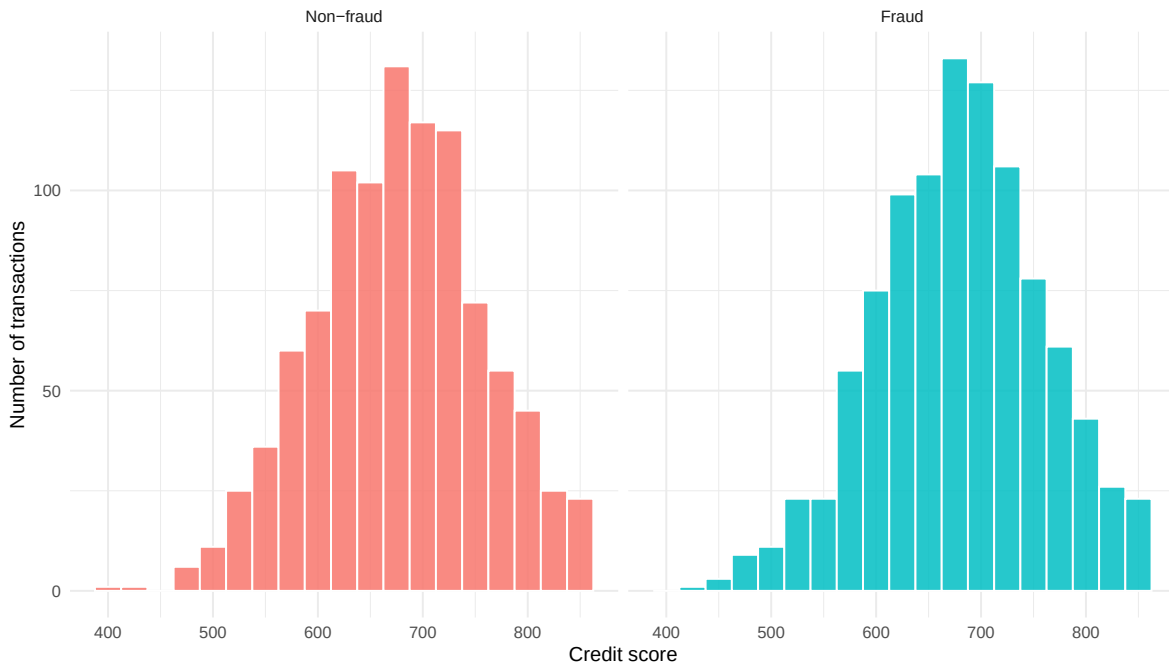


Figure 2: Distribution of credit score by fraud status in the balanced analysis sample.

The credit-score distributions of fraudulent and non-fraudulent transactions are broadly similar, with considerable overlap between the two groups. No clear visual separation is observed based on credit score alone. This suggests that credit score does not appear to be a strong standalone indicator of fraud in this balanced analysis sample.

Account Age

Account age is examined as an account-profile characteristic. Comparing account-age distributions between fraud and non-fraud transactions provides a descriptive view of whether account maturity differs across the two fraud-status groups in the balanced analysis sample.

Table 3: Account-age summary by fraud status

fraud_status	sample_size	mean_age	med_age	std_age	min_age	max_age
Non-fraud	1000	4.68	3.2	4.73	0.1	30
Fraud	1000	5.02	3.6	5.06	0.1	30

The following faceted histograms compare account-age distributions between the two fraud-status groups.

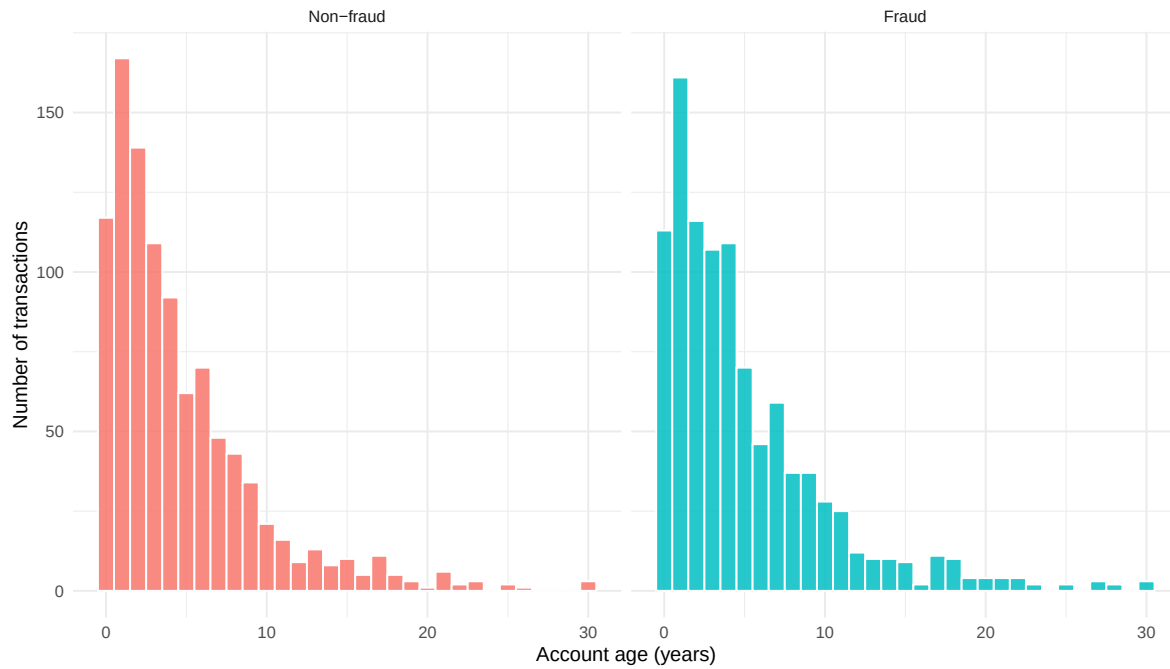


Figure 3: Distribution of account age by fraud status in the balanced analysis sample.

Overall, customer profile variables—including customer age, credit score, and account age—exhibit substantial overlap between fraud and non-fraud transactions. These variables alone do not provide clear visual discrimination between the two groups. Therefore, the following section focuses on customer behavioural characteristics, which may offer more informative patterns for exploratory analysis.

Customer Behavior Analysis

Geographic Analysis

Multivariate Segment Analysis

Key Findings

Business Insights

Conclusion